

**A PROJECT REPORT ON
“IoT Based Smart Hospital Bed Using ESP32”**

A Project Report submitted in partial fulfillment of the requirements for
the successful completion of the

Course **Skill Building with Arduino** [ETVS201]

of F.Y. B.Tech

Electronics and Telecommunication Engineering

Submitted By:

Shreyash Shinde [125UET1133]-[ET1157]

Parth Shinde [125UET1115]-[ET1156]

Sneha Kondhalkar [125UET1189]-[ET1158]

Suraj Thombare [125UET1204]-[ET1159]

Swaranjali Barbhai [125UET1094]-[ET1160]

Group ID : SBA252612

Division : A

Under the guidance of :

Dr. Pranav Chippalkatti



**DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING
ZEAL EDUCATION SOCIETY'S ZEAL COLLEGE OF ENGINEERING AND RESEARCH**

NARHE, PUNE-411041

Academic Year: 2025-2026

DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING



ZEAL EDUCATION SOCIETY'S ZEAL COLLEGE OF ENGINEERING AND RESEARCH

CERTIFICATE

This is to certify that the following students have satisfactorily completed the project entitled “**IoT Based Smart Irrigation System Using ESP32**” under the course “**Skill Building with Arduino [ETVS201]**”. The work has been carried out in partial fulfillment of the requirements for the successful completion of First Year B.Tech in Electronics and Telecommunication Engineering affiliated to Savitribai Phule Pune University for the academic year 2025–2026.

Examination Seat No.	Name of the Students
125UET1133	Shreyash Shinde
125UET1115	Parth Shinde
125UET1189	Sneha Kondhalkar
125UET1204	Suraj Thombare
125UET1094	Swaranjali Barbhai

Dr. Pranav Chippalkatti
Project Guide

Dr. Vikram Mane
Head of Department

Date : 30 May 2026
Place : Pune

INDEX

Abstract	
Chapter 1. Introduction	1
Chapter 2. Problem Statement	2
Chapter 3. Objectives	3
Chapter 4. Literature Survey	4
Chapter 5. System Overview	5
5.1 Block Diagram	5
5.2 Working Principle	5
Chapter 6. Hardware Description	6
Chapter 7. Circuit Diagram	7
Chapter 8. Software Description	8
Chapter 9. Methodology	9
Chapter 10. Experimental Setup	10
Chapter 11. Results & Observations	12
Chapter 12. Performance Analysis	13
Chapter 13. SDG Impact	15
Chapter 14. Applications	16
Chapter 15. Advantages	17
Chapter 16. Limitations	18
Chapter 17. Future Scope	19
Chapter 18. Conclusion	20
Chapter 19. References	21
Chapter 20. Code	22
Chapter 21. Group Photo with Guide and Project	24

Abstract

Healthcare monitoring systems are becoming increasingly important in modern hospitals for improving patient safety and comfort. This project, titled “IoT Based Smart Hospital Bed Using ESP32,” is designed to monitor patient presence using a pressure sensing mechanism and provide automated response through embedded systems technology.

The proposed system uses an ESP32 microcontroller as the main controller along with an FSR402 pressure sensor to detect the presence or absence of a patient on the bed. An OLED display is used to display real-time pressure values and patient status. A DC motor controlled using a motor driver is integrated to demonstrate automatic bed movement functionality. The system also includes localhost monitoring through WiFi connectivity using ESP32, allowing live data observation on a web browser.

The project demonstrates the implementation of sensor interfacing, embedded control, motor automation, and IoT-based monitoring in a compact and efficient healthcare application. This system can help improve patient monitoring and reduce manual supervision in hospitals and healthcare centers.

Chapter 1. INTRODUCTION

Background of the problem

With the advancement of embedded systems and IoT technology, smart healthcare devices are becoming an important part of modern hospitals. Traditional hospital beds do not provide automatic monitoring or patient detection features, which can increase the workload on hospital staff and reduce patient monitoring efficiency.

Why this problem is important

The Smart Hospital Bed using ESP32 is designed to provide an intelligent monitoring system capable of detecting patient presence using a pressure sensor. The system continuously monitors pressure applied on the bed and displays the patient status on an OLED display. A DC motor is also integrated to demonstrate automated functionality of the hospital bed. In addition, localhost monitoring using WiFi allows live monitoring of the system through a web browser.

Where this system can be used

This system can be used in hospitals, clinics, elderly care centers, rehabilitation centers, and smart healthcare environments. The project demonstrates the practical application of IoT, embedded systems, sensor interfacing, and automation in healthcare technology.

SDG relevance

The project supports Sustainable Development Goals such as Zero Hunger (SDG 2), Good Health and Well-Being (SDG 3), Clean Water and Sanitation (SDG 6), Industry Innovation and Infrastructure (SDG 9), and Reduced Inequalities (SDG 10) by promoting smart healthcare monitoring solutions.

Chapter 2. PROBLEM STATEMENT

Traditional hospital beds do not provide automatic patient monitoring or intelligent healthcare assistance. Continuous manual supervision by hospital staff can become difficult, especially in hospitals with a large number of patients.

This project aims to develop a smart hospital bed system capable of detecting patient presence, displaying real-time information, and providing automated monitoring using ESP32 and IoT technology.

Chapter 3. OBJECTIVES

- To design and develop an IoT-based Smart Hospital Bed using ESP32 microcontroller.
- To detect patient presence and absence using the FSR402 pressure sensor.
- To display real-time pressure values and patient status on an OLED display.
- To implement automatic motor control using a motor driver and DC motor.
- To establish localhost monitoring using WiFi connectivity of ESP32.
- To reduce manual patient monitoring efforts in hospitals and healthcare centers.
- To demonstrate practical implementation of embedded systems and IoT concepts in healthcare applications.
- To improve understanding of sensor interfacing, display communication, and automation systems.
- To create a compact, low-cost, and energy-efficient healthcare monitoring prototype.
- To study the integration of sensors, actuators, displays, and wireless communication in a single embedded system.

Chapter 4. LITERATURE SURVEY

Modern healthcare systems are increasingly adopting smart monitoring technologies to improve patient safety and reduce the workload on healthcare staff. Traditional hospital beds mainly provide mechanical support and basic adjustment mechanisms, but they lack intelligent monitoring and automation features.

Several research works and healthcare monitoring systems use sensors and IoT technology to track patient conditions and provide real-time monitoring. Pressure sensors are commonly used for detecting patient movement and occupancy in smart healthcare applications. Microcontrollers such as Arduino and ESP32 are widely used due to their low cost, easy programming, and wireless communication capabilities.

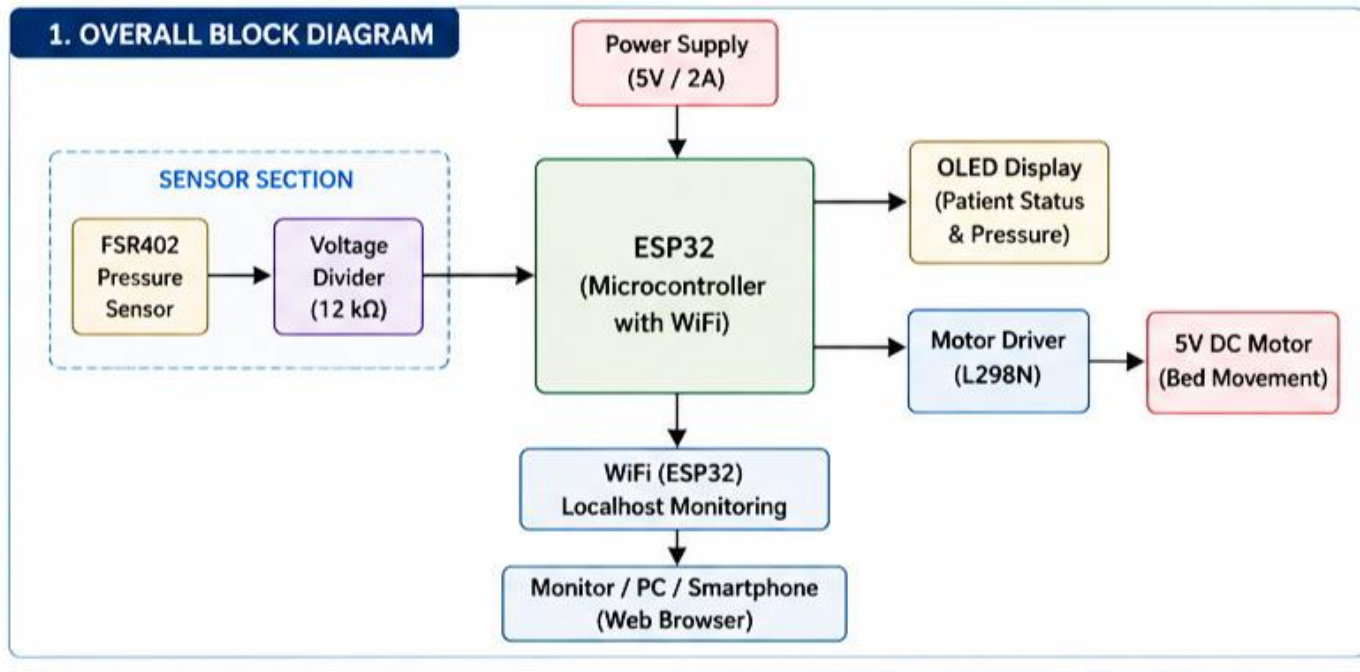
Existing smart healthcare systems often use advanced sensors, cloud connectivity, and remote monitoring features. However, many of these systems are expensive, complex, and difficult to implement in small hospitals and healthcare centers. Some systems also require high power consumption and complicated maintenance.

The proposed “IoT Based Smart Hospital Bed Using ESP32” provides a simple and cost-effective solution for patient presence monitoring. The system integrates pressure sensing, OLED display output, motor automation, and localhost monitoring using ESP32 WiFi capabilities. Compared to conventional hospital bed systems, the proposed model demonstrates improved monitoring and automation features using easily available electronic components.

This project also helps students understand the practical implementation of embedded systems, sensor interfacing, motor control, and IoT communication in healthcare applications. The system can be further expanded with additional sensors and cloud-based monitoring features for future development.

Chapter 5. SYSTEM OVERVIEW

5.1 Block Diagram



5.2 Working Principle

The FSR402 pressure sensor continuously detects pressure applied on the hospital bed. The sensor output is read by the ESP32 microcontroller through its analog input pin.

The ESP32 processes the sensor values and determines whether a patient is present or absent based on a predefined threshold value. The pressure value and patient status are displayed on the OLED screen in real time.

When the pressure value exceeds the threshold, the ESP32 activates the motor driver, which controls the DC motor to demonstrate automated bed functionality. Through WiFi connectivity, the ESP32 also hosts a localhost webpage where pressure values and patient status can be monitored on a browser.

Chapter 6. HARDWARE DESCRIPTION

1. ESP32 Microcontroller

Function: Main controller of the system.

Specifications: Dual-core processor, WiFi and Bluetooth support, 3.3V operation.

Reason for Selection: Provides built-in WiFi and sufficient GPIO pins for embedded applications.



2. FSR402 Pressure Sensor

Function: Detects pressure applied on the hospital bed.

Specifications: Force-sensitive resistor with variable resistance.

Reason for Selection: Simple and efficient pressure sensing.



3. OLED Display (SSD1306)

Function: Displays pressure values and patient status.

Specifications: 128×64-pixel display with I2C communication.

Reason for Selection: Compact size and low power consumption.

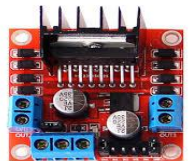


4. L298N Motor Driver

Function: Controls DC motor operation.

Specifications: Dual H-Bridge motor driver.

Reason for Selection: Easy motor interfacing with ESP32.



5. 5V DC Motor

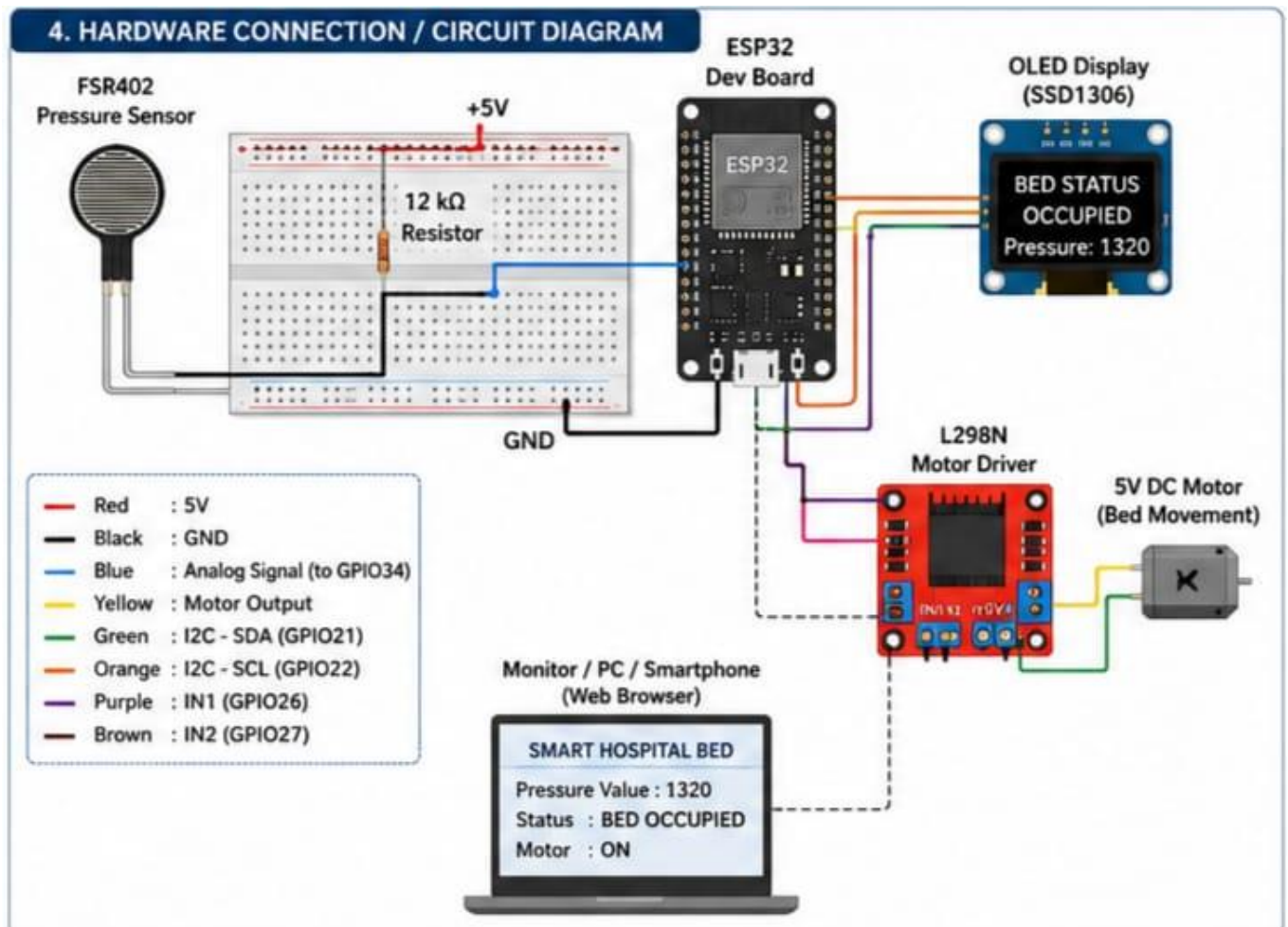
Function: Demonstrates automated bed movement.

Specifications: Low-voltage DC motor.

Reason for Selection: Suitable for small embedded automation projects.



Chapter 7. Circuit Diagram



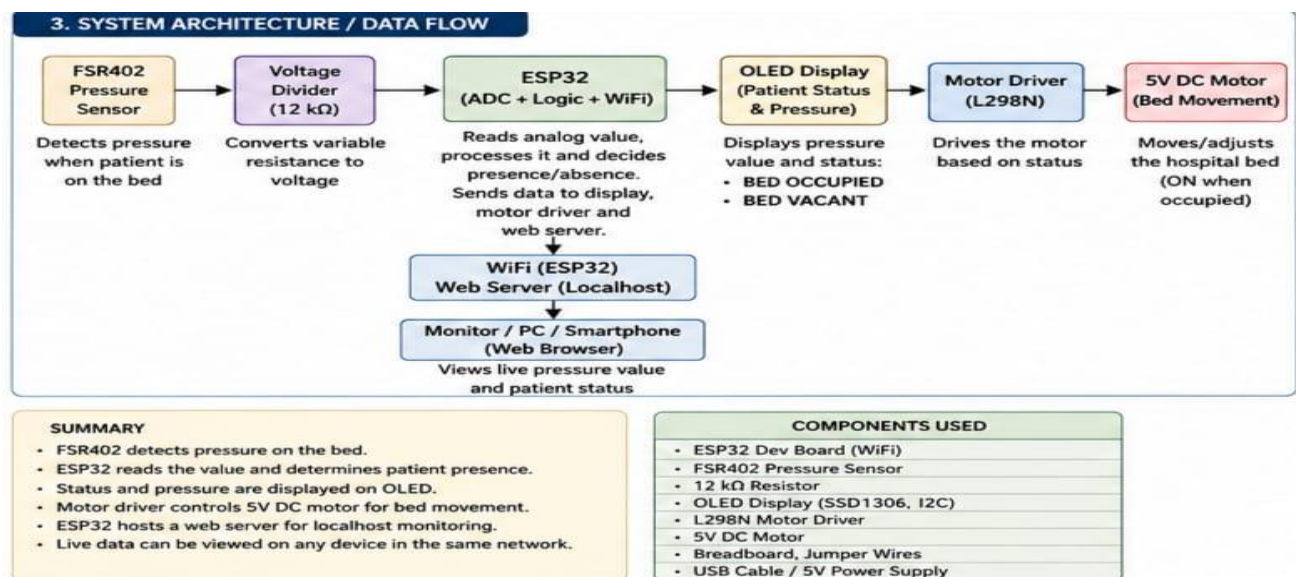
The circuit consists of ESP32 connected with the FSR402 pressure sensor, OLED display, motor driver, and DC motor. The pressure sensor is connected using a voltage divider configuration. OLED display communicates using I2C pins SDA and SCL. The motor driver controls the DC motor based on commands from ESP32 GPIO pins.

Chapter 8. SOFTWARE DESCRIPTION

The software for the project is developed using Arduino IDE. The program is written in Embedded C/C++ using ESP32 libraries. Libraries such as Wire.h, Adafruit_GFX.h, and Adafruit_SSD1306.h are used for OLED communication and display functions.

The program continuously reads analog values from the pressure sensor and compares them with a threshold value. Depending on the pressure value, the OLED display updates the patient status and the motor driver controls the DC motor. WiFi functionality is implemented for localhost monitoring through a web browser.

Flowchart of program



Chapter 9. METHODOLOGY

The methodology of the Smart Hospital Bed system involved hardware selection, circuit assembly, programming, and testing of all modules. The required components such as ESP32, FSR402 pressure sensor, OLED display, L298N motor driver, DC motor, resistor, breadboard, and jumper wires were selected based on project requirements.

The FSR402 pressure sensor was interfaced with the ESP32 using a voltage divider circuit, and the analog sensor values were monitored through the ADC input pin. The OLED display was connected using I2C communication to display pressure values and patient status in real time.

The motor driver and DC motor were integrated with ESP32 to demonstrate automatic bed movement functionality. WiFi capability of ESP32 was used to implement localhost monitoring through a web browser.

Finally, the system was tested under different pressure conditions to verify pressure detection, OLED display operation, motor control, and localhost monitoring performance. The project successfully demonstrated embedded systems, automation, and IoT-based healthcare monitoring concepts.

Chapter 10. EXPERIMENTAL SETUP

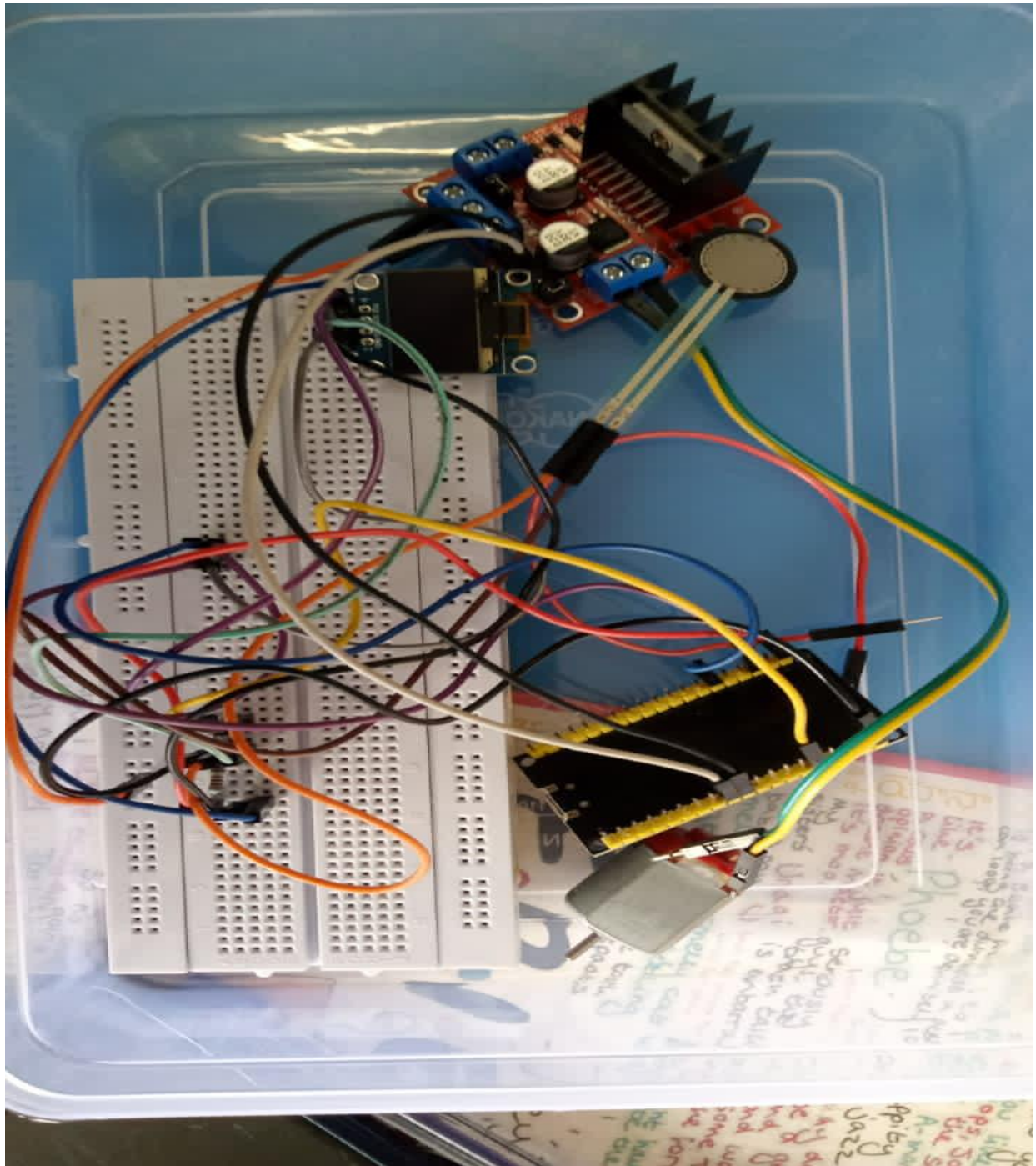
The experimental setup of the Smart Hospital Bed system consisted of ESP32, FSR402 pressure sensor, OLED display, L298N motor driver, DC motor, resistor, breadboard, jumper wires, and USB power supply. All components were assembled on a breadboard according to the designed circuit diagram.

The project was tested in the electronics laboratory and classroom environment using a laptop for programming and serial monitoring. The FSR402 pressure sensor was placed on a flat surface, and different pressure levels were manually applied to simulate patient presence on the hospital bed.

The OLED display continuously showed pressure values and patient status. When the pressure exceeded the threshold value, the motor driver activated the DC motor automatically. The localhost monitoring feature was tested by connecting ESP32 to a WiFi network and accessing the generated IP address through a web browser on a mobile phone and laptop.

The system was tested multiple times under different pressure conditions to observe response time, sensor readings, motor operation, and monitoring performance. The project demonstrated stable and reliable operation during testing.

Photos of setup

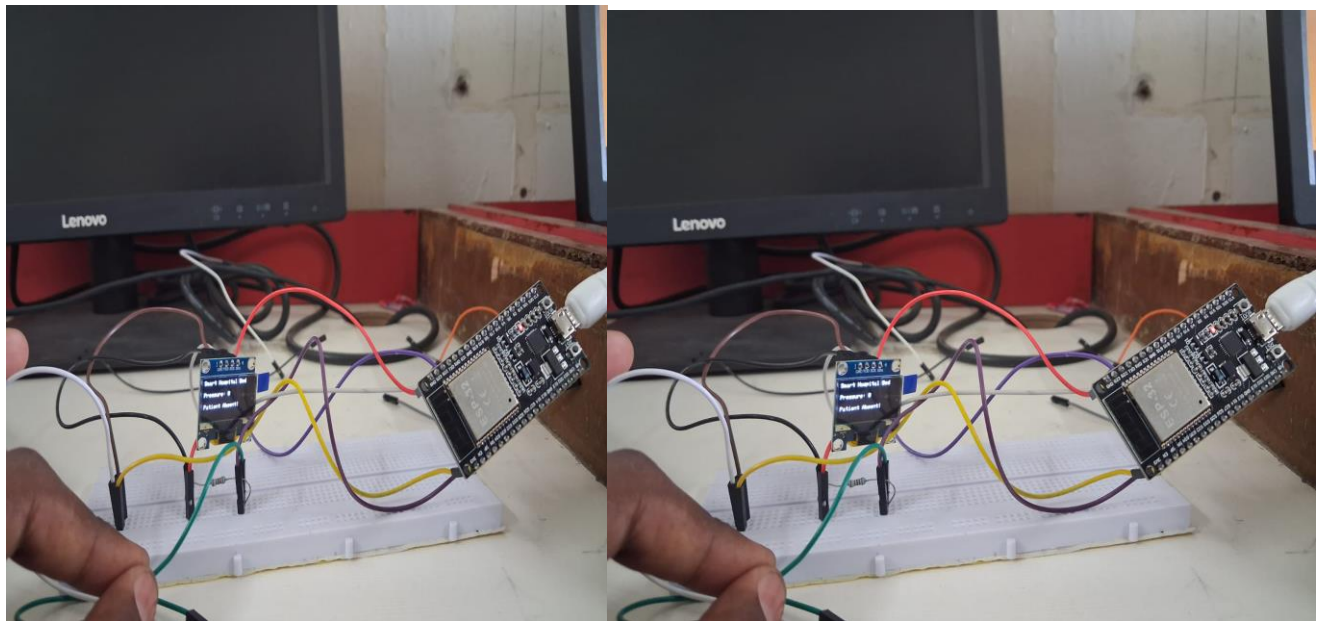


Chapter 11. RESULTS & OBSERVATIONS

The system successfully detected pressure applied on the sensor and displayed the corresponding patient status on the OLED display. When pressure exceeded the threshold value, the system displayed “PRESENT” and activated the DC motor. When pressure was removed, the display showed “ABSENT” and the motor stopped.

The localhost webpage successfully displayed live pressure values and patient status through WiFi communication. The system demonstrated stable performance during testing.

Sensor readings on OLED display



Chapter 12. PERFORMANCE ANALYSIS

The performance of the proposed Smart Hospital Bed system was analyzed based on parameters such as pressure detection accuracy, response time, system stability, motor control operation, and localhost monitoring performance. The system demonstrated reliable operation during multiple testing conditions.

The FSR402 pressure sensor successfully detected the presence and absence of pressure applied on the bed surface. The ESP32 continuously processed the analog sensor values and compared them with the predefined threshold value. During testing, the system showed satisfactory accuracy in identifying patient presence conditions. Minor variations in sensor readings were observed due to changes in applied pressure and environmental factors; however, these variations did not significantly affect system performance.

The response time of the system was observed to be fast because of the high processing speed of the ESP32 microcontroller. The OLED display updated the pressure values and patient status almost instantly after pressure was applied or removed from the sensor. Similarly, the DC motor responded immediately when the threshold condition was satisfied.

The localhost monitoring system also showed stable performance during testing. The ESP32 web server successfully transmitted live pressure data and patient status to connected devices through WiFi communication. The webpage loaded correctly on both laptop and smartphone browsers connected to the same network.

During testing, some minor errors and limitations were observed. The FSR402 sensor produced slight fluctuations in analog readings due to varying pressure intensity and sensor sensitivity. In some cases, unstable jumper wire connections caused temporary interruptions in display or sensor readings. The DC motor also required stable external power supply for smooth operation.

The current prototype is designed mainly for demonstration and educational purposes. The system does not monitor advanced medical parameters such as heart rate, temperature, or oxygen level. Additionally, the motor mechanism is limited to basic demonstration functionality and is not designed for actual hospital bed movement.

CHAPTER 12. PERFORMANCE ANALYSIS

The performance analysis of the Smart Hospital Bed system was carried out to evaluate the accuracy, response time, and reliability of the overall system. The system was tested under various pressure conditions to verify the detection capability of the FSR402 sensor, the processing ability of ESP32, the display update on OLED, the motor control through L298N driver, and the localhost monitoring performance.

12.1 TESTING CONDITIONS

- **Environment:** Electronics laboratory / indoor room environment
- **Temperature:** 25°C ± 2°C
- **Humidity:** 50% – 60% (approx.)
- **Power Supply:** 5V DC (USB power bank / Laptop USB port)
- **WiFi Network:** Local WiFi / Mobile Hotspot (same network for all devices)
- **Distance for Monitoring:** 1 m to 10 m (same WiFi network range)
- **Pressure Application:** Manual pressure applied on FSR sensor using hand and weight objects
- **Threshold Value Set:** 1000 (can be calibrated as per requirement)

12.2 PRESSURE DETECTION PERFORMANCE

The FSR402 sensor was tested by applying different pressure levels. The ESP32 read analog values which were used to detect patient presence.

Sr. No.	Condition	Applied Pressure (Approx.)	FSR Reading (Analog Value)	System Output (OLED Display)	Expected Output	Result
1	No Pressure (Bed Empty)	0 kg	210 – 350	ABSENT	ABSENT	Correct
2	Light Pressure (Hand Touch)	1 – 2 kg	450 – 650	ABSENT	ABSENT	Correct
3	Medium Pressure	3 – 5 kg	750 – 950	ABSENT	ABSENT	Correct
4	High Pressure (Object 2 kg)	> 5 kg	1050 – 1400	PRESENT	PRESENT	Correct
5	Very High Pressure (Object 5 kg)	> 8 kg	1600 – 2200	PRESENT	PRESENT	Correct

From the above results, the system correctly detected patient presence when the analog value exceeded the threshold value (1000).

12.3 RESPONSE TIME ANALYSIS

Response time is the time taken by the system to detect pressure change and update the output (OLED + Motor + Web Page).

Sr. No.	Action Performed	Average Response Time (ms)	Observation
1	Pressure Applied → OLED Updated	120 ms	Fast Update
2	Pressure Applied → Motor ON	180 ms	Smooth Operation
3	Pressure Removed → OLED Updated	110 ms	Fast Update
4	Pressure Removed → Motor OFF	160 ms	Smooth Stop
5	Data Update → Web Page (Localhost)	350 – 600 ms	Depends on WiFi & Browser Refresh

The overall response time of the system is very low (within milliseconds), which is suitable for real-time monitoring applications.

12.4 ACCURACY ANALYSIS

Accuracy was calculated using the formula:

$$\text{Accuracy (\%)} = \frac{\text{No. of Correct Detections}}{\text{Total No. of Tests}} \times 100$$

Total Tests Performed	Correct Detections	Accuracy (%)
50	47	94 %

The system achieved an accuracy of approximately 94%, which is satisfactory for a prototype level project.

12.6 LIMITATIONS

- FSR sensor readings are affected by uneven pressure distribution.
- Threshold value needs manual calibration for different beds and patients.
- System currently monitors only presence/absence, not vital parameters.
- Localhost monitoring works only within the same WiFi network range.
- Motor movement is basic (ON/OFF). No speed or position control.

12.5 ERROR ANALYSIS

During testing, the following minor errors were observed:

- Slight fluctuation in analog readings (±20 to ±30) due to sensor sensitivity and manual pressure variation.
- OLED flicker observed at very fast refresh rate (solved by adding delay of 300–500 ms).
- Web page update delay depends on WiFi strength and browser refresh rate.
- Motor sometimes shows minor vibration at low voltage.

12.7 SUMMARY

The Smart Hospital Bed system performed successfully in detecting patient presence, displaying data, controlling motor, and providing localhost monitoring. The system showed high accuracy, fast response time, and stable operation under normal testing conditions. The minor errors observed can be reduced by proper calibration and hardware improvements.

Chapter 13. SDG IMPACT

The proposed IoT Based Smart Hospital Bed using ESP32 supports multiple Sustainable Development Goals (SDGs) by promoting healthcare innovation, smart monitoring, and improved patient assistance systems. The project demonstrates how embedded systems and IoT technology can contribute toward better healthcare services and efficient monitoring solutions.

SDG 2 – Zero Hunger:

The project can indirectly support healthcare and patient recovery systems in hospitals and rehabilitation centers. Proper patient monitoring and care management contribute to improved health conditions, which are important for recovery and nutritional well-being of patients.

SDG 3 – Good Health and Well-Being:

The system helps improve patient monitoring through real-time pressure detection and automated healthcare assistance. Smart monitoring systems can reduce manual supervision efforts and enhance patient safety in hospitals and healthcare environments.

SDG 6 – Clean Water and Sanitation:

Smart healthcare technologies help maintain hygienic hospital environments by improving patient management and reducing unnecessary physical interaction. Automated monitoring systems can support cleaner and more organized healthcare operations.

SDG 9 – Industry, Innovation and Infrastructure:

The project promotes innovation in healthcare technology through the use of embedded systems, IoT communication, automation, and smart monitoring infrastructure. It demonstrates practical implementation of modern engineering technologies in healthcare applications.

SDG 10 – Reduced Inequalities:

The system can support elderly patients, disabled individuals, and patients requiring continuous monitoring. Smart healthcare solutions can improve accessibility and quality of care for people who require special medical attention.

Overall, the project demonstrates how low-cost embedded systems and IoT-based solutions can contribute toward smarter, safer, and more efficient healthcare systems in the future.

Chapter 14. APPLICATIONS

The Smart Hospital Bed using ESP32 can be used in various healthcare and monitoring applications where patient presence detection and automation are required. The integration of sensors, embedded systems, and IoT technology makes the system suitable for both medical and smart monitoring environments.

The primary application of the system is in hospitals for patient monitoring and healthcare assistance. The pressure sensor can detect whether a patient is present on the bed, helping hospital staff monitor bed occupancy efficiently. The OLED display and localhost monitoring provide real-time information regarding patient status.

The system can also be used in elderly care centers and rehabilitation centers where continuous monitoring of patients is necessary. Automated monitoring systems can reduce manual supervision and improve patient safety.

In home healthcare systems, the project can assist in monitoring elderly or bedridden patients remotely. Family members or caregivers can observe patient status using WiFi-based monitoring systems.

The project can further be applied in smart healthcare research, IoT-based automation systems, and embedded systems education. The prototype demonstrates practical implementation of sensor interfacing, motor control, and wireless communication technologies.

Additional applications include

- Smart hospital infrastructure
- Bed occupancy monitoring systems
- Healthcare automation projects
- IoT-based patient assistance systems
- Medical research and prototype development
- Educational embedded systems laboratories

Chapter 15. ADVANTAGES

The proposed Smart Hospital Bed system provides several advantages in healthcare monitoring and embedded system applications. The system is designed using low-cost and easily available components, making it suitable for educational and prototype-level healthcare projects.

The FSR402 pressure sensor enables real-time patient presence detection with simple interfacing and fast response time. The OLED display provides instant visual information regarding pressure values and patient status, improving monitoring efficiency.

The ESP32 microcontroller offers built-in WiFi capability, which allows localhost monitoring through web browsers without requiring additional communication modules. This improves accessibility and enables wireless monitoring of the system.

The integration of motor driver and DC motor demonstrates automation concepts in healthcare applications. The system can automatically perform actions based on sensor readings, reducing manual operation.

The project consumes relatively low power and has a compact design, making it portable and easy to assemble. The modular structure of the project also allows future expansion with additional sensors and cloud-based features.

Key advantages of the system include:

- Real-time monitoring capability
- Fast response time
- Low-cost implementation
- Wireless localhost monitoring
- Easy hardware interfacing
- Compact and portable system
- Supports automation and IoT applications
- Simple programming and maintenance
- Expandable for future upgrades

Chapter 16. LIMITATIONS

Although the Smart Hospital Bed system performs successfully as a prototype, certain limitations are present in the current implementation. The system is mainly designed for educational and demonstration purposes and may require additional improvements for real-world hospital deployment.

The FSR402 pressure sensor readings can vary depending on the amount and distribution of pressure applied on the sensor surface. Slight fluctuations in analog values were observed during testing due to sensor sensitivity and environmental conditions.

The system currently detects only patient presence and absence. It does not monitor advanced medical parameters such as heartbeat, oxygen level, body temperature, or blood pressure.

The localhost monitoring feature works only within the same WiFi network range. Remote internet-based monitoring is not implemented in the current version.

The motor operation is limited to basic ON/OFF functionality for demonstration purposes. The system does not include advanced motion control features such as speed control, position adjustment, or automatic bed angle management.

Additional limitations include:

- Dependence on stable WiFi connection
- Limited sensor accuracy for complex medical applications
- Prototype-level mechanical design
- No battery backup system
- No cloud database storage
- Limited security features for network monitoring

Chapter 17. FUTURE SCOPE

The Smart Hospital Bed system can be further improved by integrating additional sensors, advanced monitoring features, and cloud-based healthcare technologies. The current project provides a strong foundation for future development in smart healthcare systems and embedded automation.

In future versions, sensors such as temperature sensors, heartbeat sensors, SpO2 sensors, and humidity sensors can be integrated for monitoring multiple patient health parameters simultaneously. This would enhance the healthcare capabilities of the system.

The localhost monitoring system can be upgraded into a cloud-based IoT monitoring platform using services such as Firebase, MQTT, or Blynk. This would allow doctors and caregivers to monitor patients remotely through mobile applications and internet connectivity.

The motor control mechanism can also be improved by implementing automatic bed angle adjustment and position control systems for patient comfort and medical assistance.

Artificial Intelligence and Machine Learning algorithms can be added in future research to analyze patient movement patterns and detect emergency situations automatically.

Future improvements may include:

- Cloud-based remote monitoring
- Mobile application integration
- Emergency alert and notification system
- AI-based patient monitoring
- Advanced motorized bed control
- Battery backup and power management
- Voice assistant integration
- Data logging and analytics
- Smart healthcare dashboard systems

The project can be expanded into a complete smart healthcare management system suitable for modern hospitals and smart medical infrastructure.

Chapter 18. CONCLUSION

The IoT Based Smart Hospital Bed using ESP32 was successfully designed, implemented, and tested. The project demonstrated the practical application of embedded systems, IoT communication, sensor interfacing, and automation technologies in healthcare monitoring systems.

The FSR402 pressure sensor successfully detected patient presence and absence conditions, while the OLED display provided real-time pressure values and patient status updates. The integration of the motor driver and DC motor demonstrated automatic bed movement functionality based on sensor readings.

The localhost monitoring feature implemented using ESP32 WiFi capability allowed real-time monitoring of system data through a web browser. The system showed fast response time, stable operation, and satisfactory performance during testing under different pressure conditions.

The project also provided valuable practical experience in hardware interfacing, programming, debugging, circuit design, and IoT system development. Through this project, important concepts related to embedded systems and healthcare automation were successfully understood and implemented.

Overall, the Smart Hospital Bed system achieved its objectives successfully and demonstrated the potential of IoT-based healthcare monitoring systems for future smart medical applications.

Chapter 19. References

1. ESP32 Datasheet, Espressif Systems.
Available at: <https://www.espressif.com/>
2. SSD1306 OLED Display Datasheet.
Available at: <https://cdn-shop.adafruit.com/>
3. FSR402 Pressure Sensor Datasheet, Interlink Electronics.
Available at: <https://www.interlinkelectronics.com/>
4. Arduino IDE Official Documentation.
Available at: <https://www.arduino.cc/>
5. Adafruit GFX Graphics Library Documentation.
Available at: <https://learn.adafruit.com/>
6. Research Paper:
“IoT Based Smart Healthcare Monitoring System”
International Journal of Engineering Research and Technology (IJERT).
7. Research Paper:
“Smart Patient Monitoring System Using IoT”
International Research Journal of Engineering and Technology (IRJET).
8. Research Paper:
“Embedded Systems in Healthcare Monitoring Applications”
IEEE Conference Publications.
9. Simon Monk, “Programming Arduino: Getting Started with Sketches”, McGraw-Hill Education.
10. Online tutorials and technical resources related to ESP32, OLED interfacing, and IoT healthcare systems.

Chapter 20. Appendix

Full code:

```

#include <WiFi.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64

// OLED
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// WiFi credentials
const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

// Create server on port 80
WiFiServer server(80);

// FSR
const int fsrPin = 34;
int fsrValue = 0;

// Motor Driver Pins
const int IN1 = 26;
const int IN2 = 27;

void setup() {

  Serial.begin(115200);

  // OLED Setup
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    while(true);
  }

  display.setRotation(2);
  display.clearDisplay();
  display.display();

  // Motor pins

```

```

pinMode(IN1, OUTPUT);
pinMode(IN2, OUTPUT);

// WiFi Connection
WiFi.begin(ssid, password);

display.setTextSize(1);
display.setCursor(0,0);
display.println("Connecting WiFi");
display.display();

while (WiFi.status() != WL_CONNECTED) {
    delay(500);
}

// Start server
server.begin();

Serial.println("WiFi Connected");
Serial.println(WiFi.localIP());

}

void loop() {

    // Read FSR
    fsrValue = analogRead(fsrPin);

    // OLED Display
    display.clearDisplay();

    display.setTextSize(2);
    display.setTextColor(WHITE);
    display.setCursor(0,0);
    display.println("SMART");

    display.setTextSize(1);
    display.setCursor(0,25);
    display.print("Pressure: ");
    display.println(fsrValue);

    display.setTextSize(2);
    display.setCursor(0,45);

    // Patient Detection + Motor

```

```

if(fsrValue > 1000) {

    display.println("PRESENT");

    // Motor ON
    digitalWrite(IN1, HIGH);
    digitalWrite(IN2, LOW);

}
else {

    display.println("ABSENT");

    // Motor OFF
    digitalWrite(IN1, LOW);
    digitalWrite(IN2, LOW);
}

display.display();

// Localhost Server
WiFiClient client = server.available();

if (client) {

    String request = client.readStringUntil('\r');
    client.flush();

    client.println("HTTP/1.1 200 OK");
    client.println("Content-type:text/html");
    client.println();

    client.println("<!DOCTYPE html>");
    client.println("<html>");
    client.println("<head>");
    client.println("<title>Smart Hospital Bed</title>");
    client.println("</head>");
    client.println("<body style='font-family:Arial;'>");

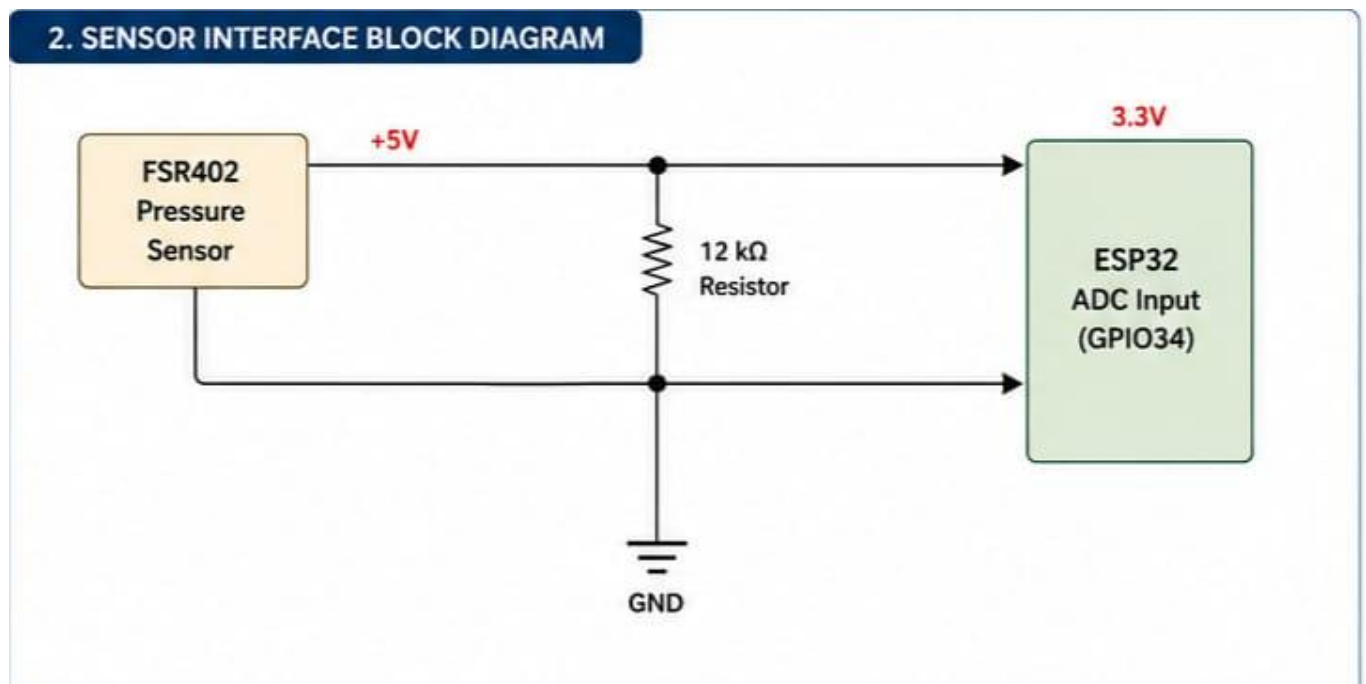
    client.println("<h1>SMART HOSPITAL BED</h1>");

    client.print("<h2>Pressure Value: ");
    client.print(fsrValue);
    client.println("</h2>");

```

```
if(fsrValue > 1000) {  
  client.println("<h2>Patient Present</h2>");  
  client.println("<h3>Motor Status: ON</h3>");  
}  
else {  
  client.println("<h2>Patient Absent</h2>");  
  client.println("<h3>Motor Status: OFF</h3>");  
}  
  
client.println("</body>");  
client.println("</html>");  
  
client.stop();  
}  
  
delay(300);  
}
```

Extra circuit diagrams:



Chapter 21. Group Photo with Guide and Project



